

Brain Tumor Detection Using Deep Learning

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A decorative graphic on the left side of the slide consists of a cluster of hexagons in various shades of blue, orange, and white. Some hexagons contain images of brains: a lateral view of a human brain, a sagittal view of a brain with a highlighted tumor, a glowing blue brain with red highlights, and a cross-section of a brain with colorful regions.

Agenda

BRAIN TUMOR
DETECTION USING DEEP
LEARNING

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Introduction

Brain tumors are a serious and potentially life-threatening medical condition that requires prompt diagnosis and treatment. However, accurately detecting brain tumors can be a challenging task for medical professionals, and misdiagnosis can have severe consequences. In recent years, deep learning has emerged as a powerful tool for medical image analysis, and brain tumor detection is one area where it has shown great promise. This project focuses on using deep learning algorithms, such as convolutional neural networks (CNNs), to aid in the detection and diagnosis of brain tumors using medical imaging data. By leveraging the power of deep learning, we aim to improve the accuracy and speed of brain tumor diagnosis, ultimately leading to better patient outcomes.



Problem Statement

The problem involves developing a CNN model that can accurately predict the presence of brain tumors in medical images and classify the tumors into different types.

The goal is to assist medical professionals in diagnosing brain tumors accurately and efficiently, potentially leading to earlier detection, improved patient outcomes, and saving lives.

Brain tumors are a leading cause of cancer-related deaths worldwide, and accurate diagnosis is critical for effective treatment.

The success of the project relies on several factors, including the quality and size of the dataset, the choice of CNN architecture and hyperparameters, and the testing and validation of the model.

Problem Solution

Develop a convolutional neural network (CNN) model that can accurately predict the presence of brain tumors in medical images and classify the tumors into different types.

Train the CNN model on a dataset of medical images of brains with and without tumors, and learn to identify patterns in the images that correspond to the presence of tumors.

Optimize the CNN architecture and hyperparameters to improve the accuracy and efficiency of the model.

Test and validate the CNN model on new, unseen medical images to ensure that it can accurately predict the presence of brain tumors and classify them into different types.

Classification of Brain Tumor

Pituitary tumor:-This type of tumor occurs in the anterior body of the pituitary gland. This is a pea-sized gland that is located behind the bridge of your nose.

Meningioma tumor:-A meningioma develops on the brain's meninges and is a tumor of leptomeningeal origin. The **meninges** are the three layers of membrane that cover the brain and spinal cord.

Glioma tumor:- Glioma is a growth of cells that starts in the brain or spinal cord. The cells in a glioma look similar to healthy brain cells called glial cells. Glial cells surround nerve cells and help them function. As a glioma grows it forms a mass of cells called a tumor.

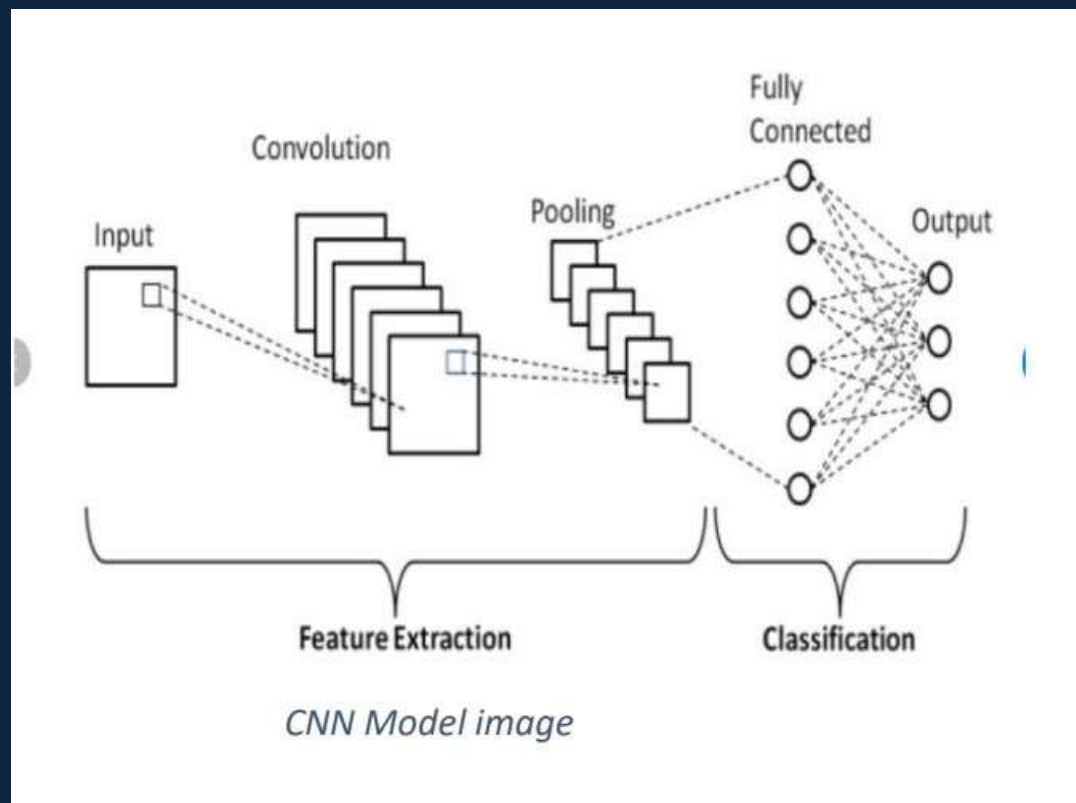
CNN model :-

One kind of deep learning model made especially for processing and analysing visual input, such photos and videos, is the convolutional neural network (CNN). It has demonstrated remarkable effectiveness in a range of computer vision applications, such as object identification, segmentation, and picture classification, owing to their inspiration from the human visual system. It is a class of deep learning models specifically designed for processing structured grid data, such as images. CNNs are capable of automatically learning and extracting relevant features from images. It is composed of multiple layers such as -

1. Convolutional layers - These layers generate a feature map by sliding a filter over the input image and recognizing patterns in images.
2. Pooling Layer - Its purpose is to reduce the size and dimensions of feature maps, which in turn makes computation faster because the number of training parameters is reduced.
3. Fully Connected Layer - Its main purpose is to connect neurons between two different layers.

Reason for using CNN model :-

- CNNs are relatively robust to variations in image data, such as changes in lighting, pose, and scale.
- CNNs excel at extracting high-level features from images, which are essential for accurate image classification.
- CNNs can be trained efficiently using specialized techniques such as stochastic gradient descent with momentum.
- CNNs are well-suited for parallel processing, which allows them to be trained on large datasets in a reasonable amount of time.
- CNNs can also be used for other tasks, such as image segmentation, object detection, and image generation.



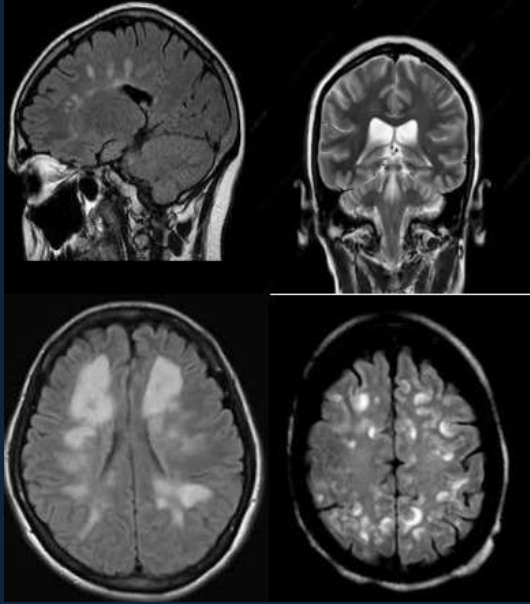
> **Input Layer:** The input layer receives the input image, which is typically represented as a three-dimensional tensor with dimensions (width, height, channels).

> **Convolutional Layer:** The convolutional layer applies a series of filters to the input image. Each filter is a small window that extracts features from the image. The output of the convolutional layer is a feature map, which contains the activations of the filters at each location in the image.

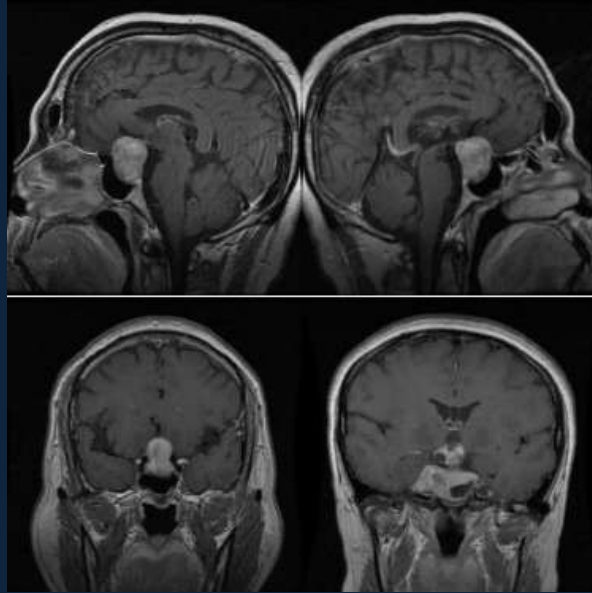
- > **Pooling Layer:** The pooling layer reduces the size of the feature map by applying a pooling operation to each region of the map. Common pooling operations include max pooling and average pooling.
- > **Fully Connected Layer:** The fully connected layer takes the flattened output of the pooling layer and transforms it into a probability distribution over the possible classes.
- > **Output Layer:** The output layer outputs the probability distribution over the possible classes. The class with the highest probability is the detected class for the input image.



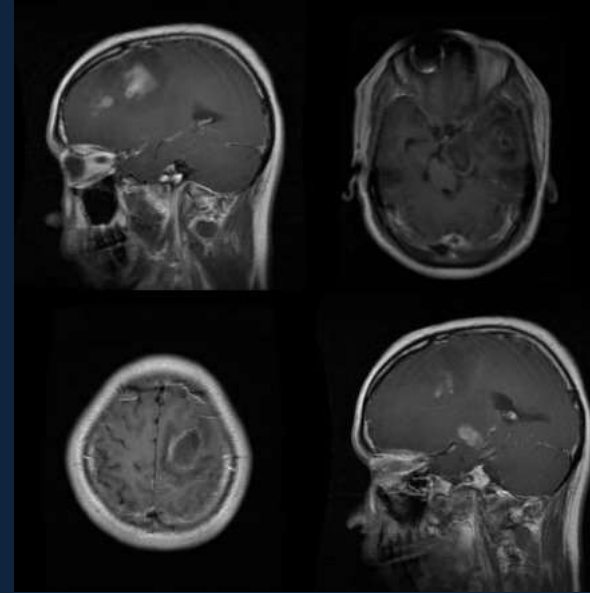
Dataset used :-



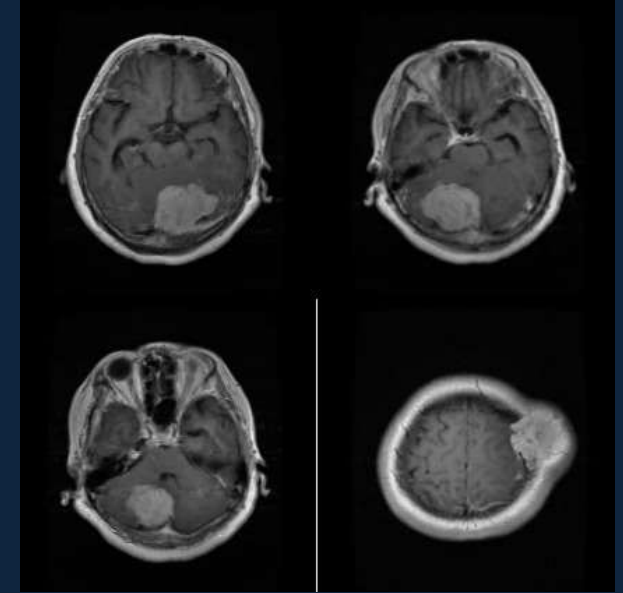
No Tumor



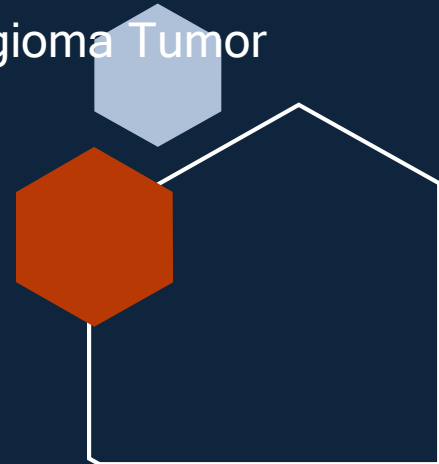
Pituitary Tumor



Glioma Tumor



Meningioma Tumor



PROJECT OUTCOME

The expected outcomes from a project on brain tumor detection using deep learning may vary depending on the specific goals and objectives of the project. However, some potential outcomes include:

- Improved accuracy and efficiency of brain tumor detection: By using deep learning algorithms, the accuracy and efficiency of brain tumor detection can be improved. This could lead to earlier detection of tumors, better treatment planning, and improved patient outcomes.
- Development of new algorithms and methods: A project in this field may involve the development of new deep learning algorithms and methods that can be used to improve brain tumor detection. These algorithms could have broader applications in medical imaging and other fields.

PROJECT OUTCOME

- Validation and evaluation of existing algorithms: A project may involve the validation and evaluation of existing deep learning algorithms for brain tumor detection. This could help to identify areas for improvement and provide insights into how these algorithms can be optimized for specific applications.
- Improved understanding of brain tumor characteristics: By analyzing large datasets of brain tumor images, a project could help to improve our understanding of the characteristics and properties of different types of brain tumors. This could lead to better treatment options and improved patient outcomes.
- Increased collaboration between medical professionals and computer scientists: A project in this field may involve collaboration between medical professionals and computer scientists, which could lead to increased knowledge sharing and interdisciplinary research. This could result in the development of new solutions and innovations that would not have been possible through traditional research methods.

COMPONENTS REQUIRED

NUMPY

NumPy module works with the numerical data. NumPy is a Python library used for working with arrays.

PANDAS

for data analysis and associated manipulation of tabular data in DataFrames.

MATPLOTLIB and SEABORN

Is used to Create a graph or we can say used for 2d visualization

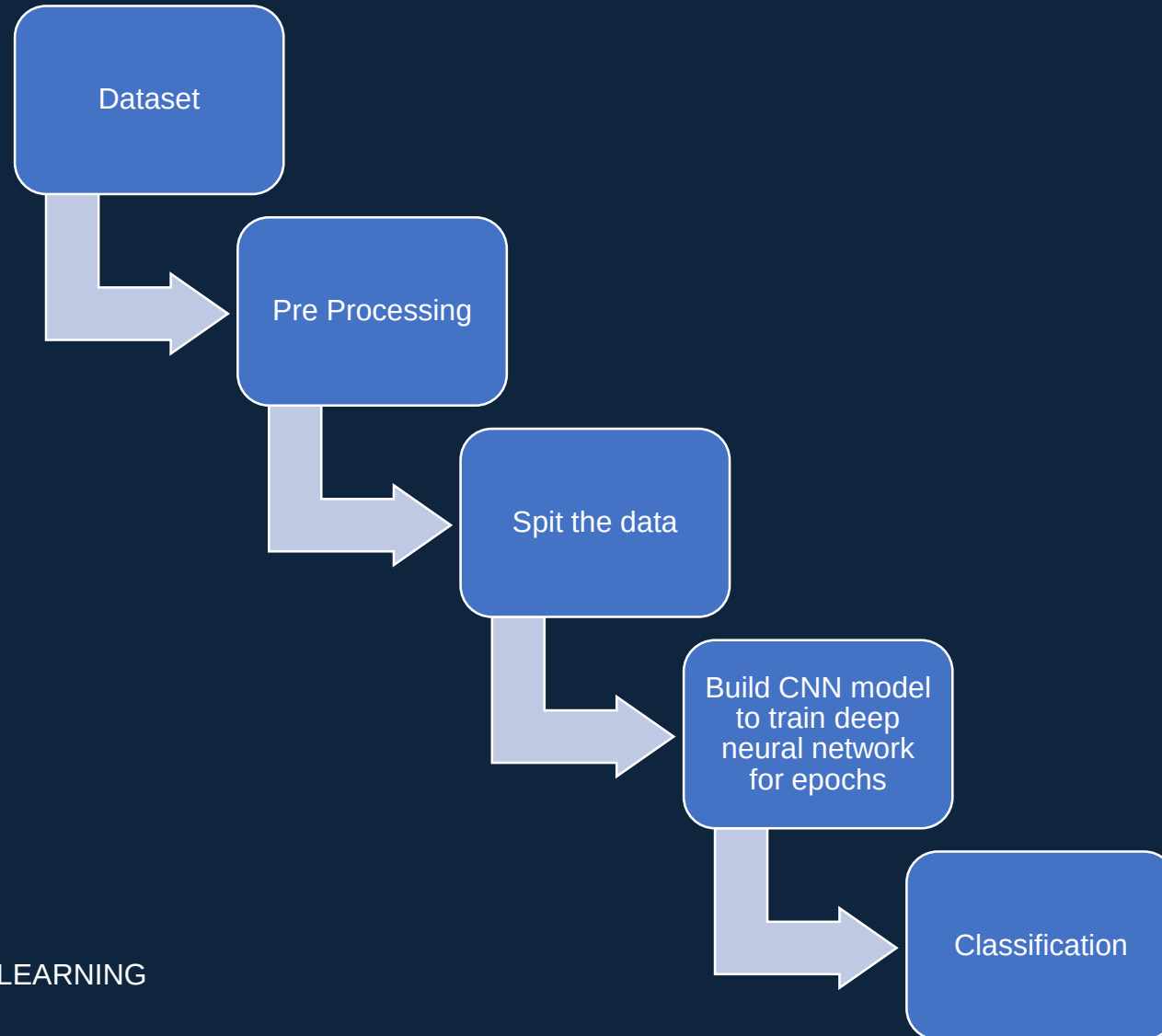
TENSORFLOW

Used to do deep learning and also develop CNN model

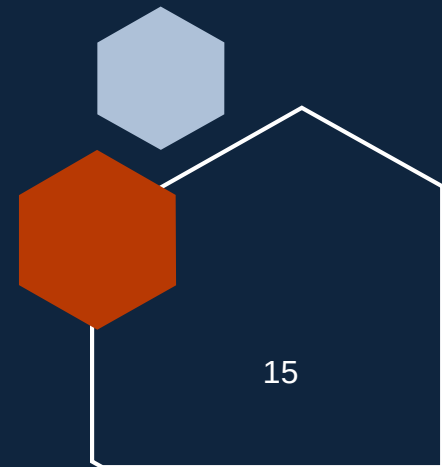
OS

provides the facility to establish the interaction between the user and the operating system

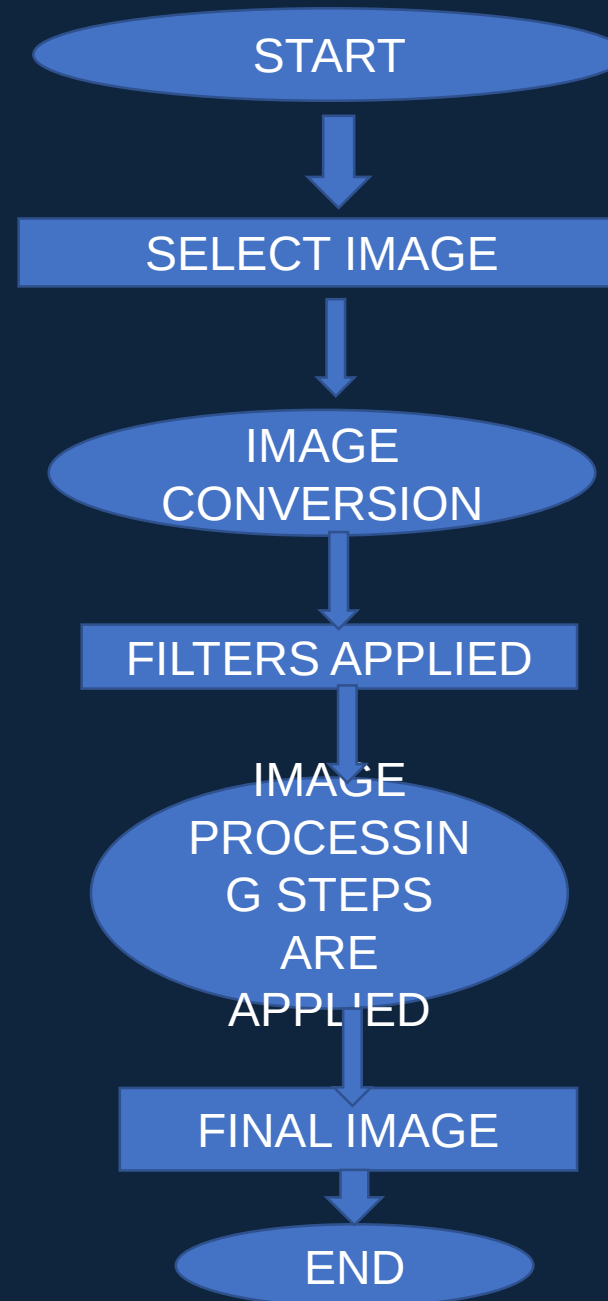
BLOCK DIAGRAM



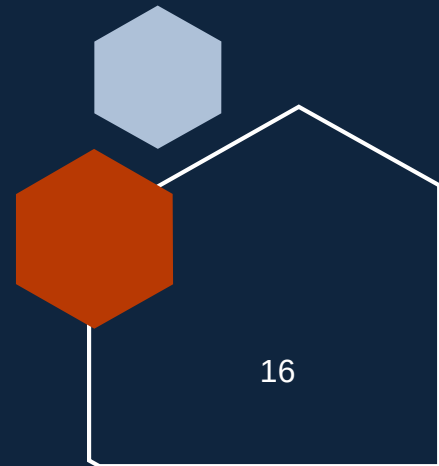
BRAIN TUMOR DETECTION USING DEEP LEARNING



FLOW CHART



BRAIN TUMOR DETECTION USING DEEP LEARNING



WORKING OF PROJECT

Data Collection: Gather a dataset of brain MRI or CT scan images, consisting of both tumor-positive and tumor-negative cases. The dataset should be appropriately labeled.

Data Preprocessing: Prepare the data for training by performing preprocessing steps such as resizing the images, normalizing pixel values, and splitting the dataset into training and testing sets.

Model Training: Train the CNN model using the training dataset. During training, the model learns to recognize patterns and features indicative of brain tumors. This is done by optimizing the model's weights and biases through an iterative process called backpropagation using a loss function and an optimizer.

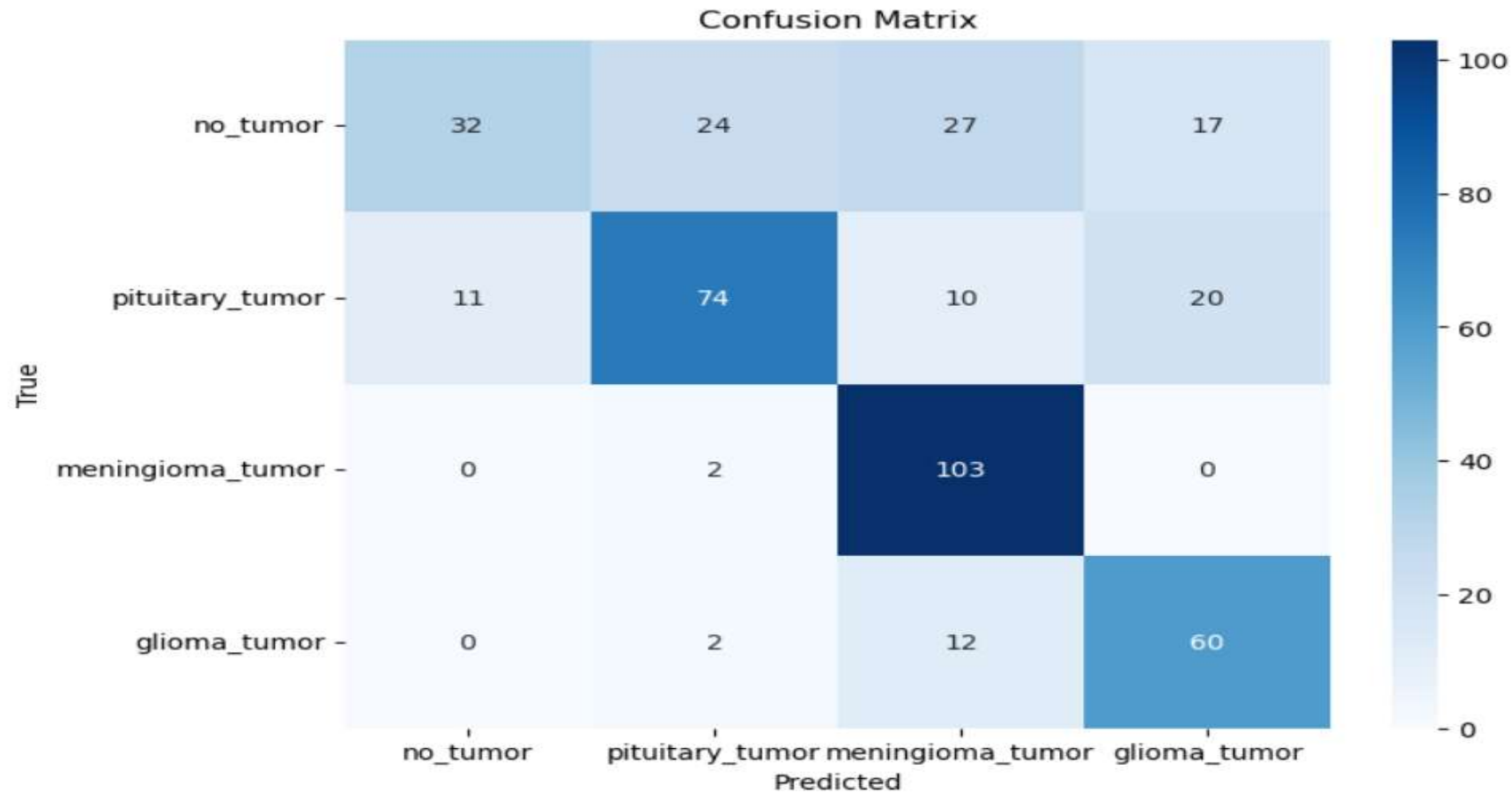
WORKING OF PROJECT

Model Evaluation: Evaluate the trained model using the testing dataset to assess its performance. Common evaluation metrics include accuracy, precision, recall. These metrics provide insights into the model's ability to correctly classify brain tumors.

Prediction and Visualization: Apply the trained model to unseen brain images to predict the presence of tumors. Visualize the predictions and compare them against ground truth labels. This step helps validate the model's effectiveness and identify any potential areas for improvement.

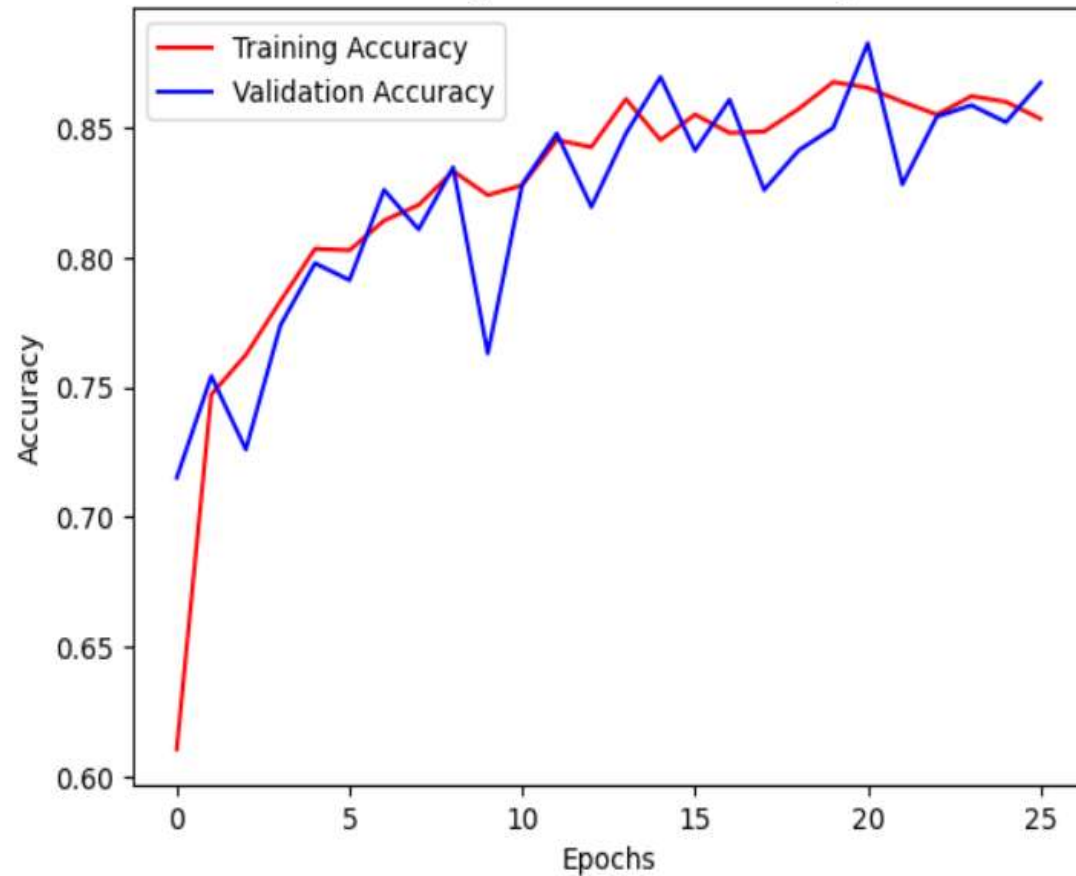
Deployment: Once the model achieves satisfactory performance, it can be deployed as a tool for assisting medical professionals in diagnosing brain tumors. The model can take new brain images as input and provide predictions or probability scores indicating the likelihood of the presence of a tumor.

Confusion Metrics

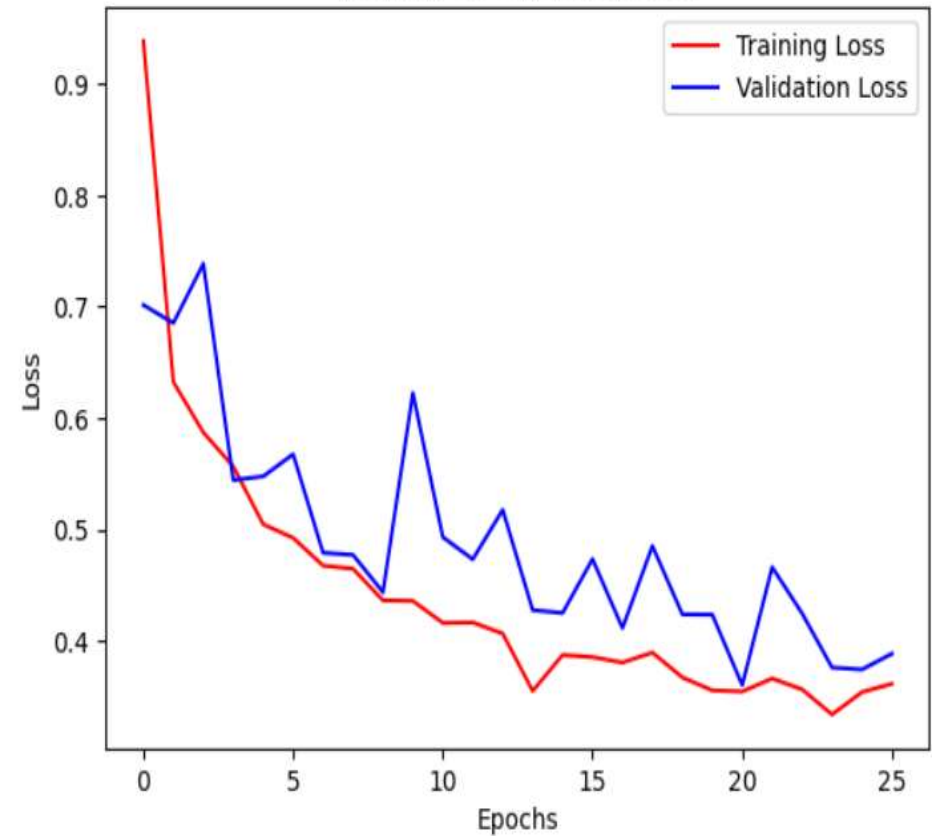


Accuracy and loss during epochs

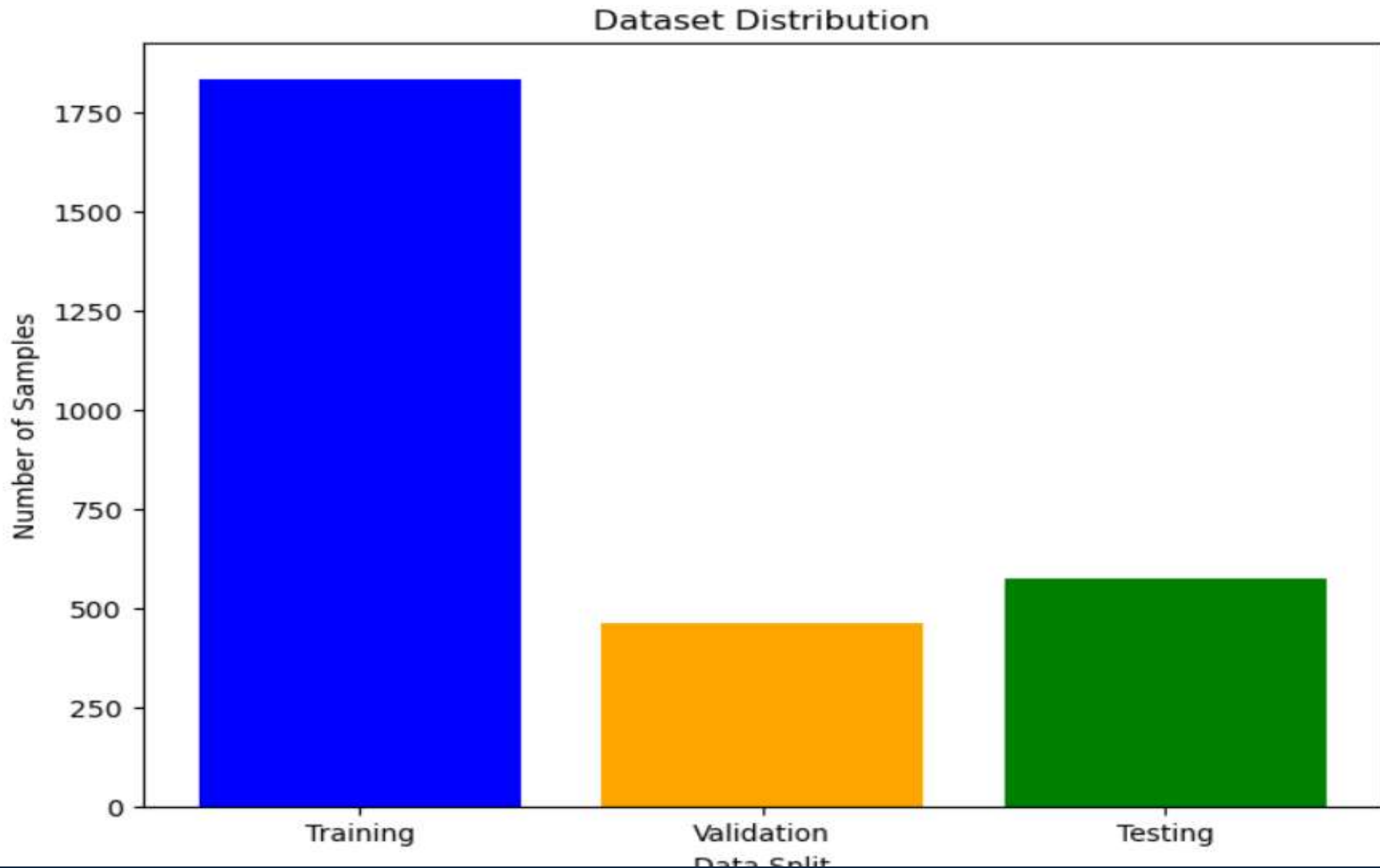
Training vs Validation Accuracy



Training vs Validation Loss



Dataset distribution



CONCLUSION

Brain tumor detection using deep learning has become a promising area of research in medical image analysis. Recent studies have demonstrated the effectiveness of deep learning-based approaches in accurately detecting and segmenting brain tumors, as well as predicting patient survival. Various CNN architectures have been utilized along with transfer learning techniques to improve model performance. However, there are still challenges and limitations that need to be addressed, such as the lack of annotated data and interpretability of deep learning models. Therefore, further research is needed to develop more robust and accurate models that can be used in clinical practice.



Thank you

